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PROJECT REPORT

Project Topic: Data-Driven Fault Detection in Process Systems Using PCA and Autoencoders

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Data-Driven Fault Detection in Process Systems Using PCA and AutoEncoders

Abstract

This project presents a data-driven approach for fault detection in process systems using AutoEncoder-based deep learning and Principal Component Analysis (PCA). The objective is to evaluate and compare the effectiveness of nonlinear and linear monitoring techniques for detecting faults in complex industrial processes. The dataset used is obtained from an offshore gas processing plant simulation and consists of 105 multivariate process variables under normal and faulty operating conditions. The AutoEncoder is trained in an unsupervised manner using normal operation data to learn the intrinsic nonlinear relationships among process variables. Fault detection is performed using reconstruction residuals, where deviations from learned patterns indicate abnormal conditions. PCA is implemented as a benchmark method using Hotelling's T^2 and Squared Prediction Error (SPE) statistics. Performance is evaluated using Fault Detection Rate (FDR), False Alarm Rate (FAR), Accuracy (ACC), and Time to Detect (TTD). The results show that the AutoEncoder achieves higher detection accuracy and faster response, particularly for nonlinear faults such as valve failures. While PCA performs adequately for faults with large variance changes, it shows limitations in capturing nonlinear process dynamics. Overall, the study demonstrates the effectiveness of nonlinear data-driven approaches for fault detection in complex systems. However, limitations such as lack of temporal modeling and sensitivity to noise are identified, suggesting the need for advanced sequence-based models in future work.

1. Introduction

Modern industrial process systems, such as offshore gas processing plants, are highly complex, nonlinear, and dynamically interconnected. Ensuring safe and reliable operation of these systems is critical, as undetected faults can lead to equipment failure, production losses, and significant safety hazards. Traditional fault detection approaches rely on accurate first-principles models; however, developing such models for large-scale industrial systems is often challenging due to process complexity and uncertainty. Data-driven methods have emerged as an effective alternative by utilizing historical process data to monitor system behavior (Venkatasubramanian et al., 2003). Among these, Principal Component Analysis (PCA) is

one of the most widely used techniques for multivariate statistical process monitoring due to its ability to capture dominant variance and correlations among variables (MacGregor & Kourti, 1995). However, PCA is fundamentally a linear method and is limited in its ability to detect faults arising from nonlinear relationships, which are prevalent in industrial processes (Venkatasubramanian et al., 2003). Recent advances in artificial intelligence and machine learning have enabled the development of nonlinear data-driven models for process monitoring. AutoEncoders, a class of deep neural networks, have demonstrated strong capability in learning complex nonlinear representations of high-dimensional data (Hinton & Salakhutdinov, 2006). These models can effectively capture intrinsic system behavior and detect subtle deviations from normal operation, making them suitable for fault detection in complex process systems (Ge et al., 2017). This project investigates the effectiveness of AutoEncoder-based fault detection and compares its performance with traditional PCA-based methods. The outcomes of this work contribute to improving process safety, reliability, and operational efficiency in industrial systems.

2. Literature review

To maintain the safety, reliability, and efficiency of Industrial systems, fault detection and diagnosis (FDD) are very crucial. Nowadays, traditional statistical techniques such as Principal Component Analysis (PCA), and modern data-driven methods such as Artificial Neural Networks (ANN), particularly Autoencoders, become very useful tools. This section reviews the literature on both approaches and highlights their comparative strengths and limitations.

2.1 Principal Component Analysis (PCA) in Fault Detection

Among all FDD methods, PCA is one of the most frequently used techniques. This technique has been widely applied in industrial and engineering applications. PCA is one appropriate technique when dealing with high-dimensional and correlated data. A principal components analysis (PCA) transforms variables from correlated variables into new uncorrelated components which represent the majority of the variability of original data and is used for dimension reduction (Gajjar et al., 2018). Therefore, this approach is effective for monitoring processes that generate large volumes of sensor data. In fault detection, one of the earliest and most frequently used applications of PCA has been to build a model using fault-free data and then look for departures from this model during the normal operation of the system. The departures are often assessed in terms of several statistical indices, such as Hotelling's T^2 statistic and the squared prediction error (SPE) or Q-statistic (Tamura & Tsujita, 2007). These statistics analyze deviations in the principal component and residual subspace, respectively, and enable fault detection when abnormal

behaviour of a system is encountered. Numerous research papers and studies have reported a variety of successful applications of PCA-based fault detection on actual systems. For example, in (Villegas et al., 2010) A pilot plant system was used to show the success of PCA-based fault detection and diagnosis techniques by comparing the values of the Q-statistic with predefined thresholds under various operating conditions, and demonstrated the ability of PCA to effectively identify the system faults even without a process physical model, purely using past operational data.

Similarly, in building energy systems, PCA has been used successfully. (Wang & Xiao, 2004) presented a PCA-based approach to the detection and diagnosis of sensor faults in building air-handling units (AHUs). Their approach was based on the Q statistic to indicate fault and contribution plots for fault diagnosis. Since the relationships in nonlinear HVAC systems are not necessarily linear, multiple PCA models were built in different operation ranges on basis of heat balance and pressure flow relations. Their results show that the PCA model can achieve high accuracy in detecting and isolating sensor faults even under conditions with fluctuating operating behavior. In the renewable energy field, PCA has also been employed for condition monitoring and fault detection. (Pozo & Vidal, 2015) presented a fault detection scheme for wind turbines using PCA and statistical hypothesis testing. A baseline PCA model was constructed from normal operational data, and it was compared with online system data in order to diagnose and detect various system faults, such as actuator and sensor failures. The study showed that the technique is effective in detecting faults in early stages, enabling the implementation of preventive maintenance actions, thus leading to reduced operational costs. On the other hand, some limitations of the application of PCA for fault detection have been identified. One limitation lies in the selection of the number of principal components. Tamura and Tsujita have indicated that the detection sensitivity to faults is significantly dependent on the choice of the number of PCs. "Inappropriate choices of the number of PCs are associated with failure to detect a fault or loss of detection sensitivity." They introduced the concept of fault signal-to-noise ratio (Fault SNR) to define an optimal number of components that would guarantee optimal fault detection performance.

The other limitations of PCA are based on the underlying assumptions of linear relation between variables, which means that PCA does not fit well with nonlinear systems. Moreover with large data sets, interpretation of the principal components might be difficult and often a component is a linear combination of a number of variables (Gajjar et al., 2018). To overcome this difficulty the notion of sparse PCA (SPCA) was introduced where, through inducing sparsity in loading vectors, components that contribute a minimum number of variables and still retain most of the variability in data, could be obtained. SPCA shows improvement over standard PCA on both the interpretation and fault diagnosis in large scale processes.

Overall, PCA has proven to be a powerful and practical tool for fault detection because of its simplicity, efficiency and ability to create a normal system behavior model using historical data. However, the success of such an approach hinges on effective model building which entails determining the right number of principal components to use and accounting for nonlinearities. These limitations have paved the way for more advanced techniques such as nonlinear PCA and other neural network based methods including autoencoders as will be discussed next.

2.2 Artificial Neural Networks and Autoencoders for Fault Detection

ANNs are increasingly popular in fault detection owing to their capacity to model the intricate nonlinear relationships present in industrial data. Traditional statistical methods struggle to uncover these intricate patterns hidden within high-dimensional and time-variant systems and, consequently, are less appropriate for modern process monitoring applications (Spina et al., 2024). The autoencoder is an ANN architecture highly effective in fault detection. This structure involves an encoder that reduces the dimensionality of the input data and a decoder that aims to reconstruct the original input. Normal operating data is employed to train the model, faults are subsequently identified by analysis of the reconstruction error, where errors greater than a normal range indicate the presence of a fault (Shang & Yu, 2024).

The primary advantage of the autoencoder over the PCA is its capability of capturing nonlinear dependencies in process data. (Fan et al., 2017) demonstrated that AE-based approaches outperform traditional methods like PCA, ICA, and various kernel-based techniques when applied to high-dimensional industrial systems such as the Tennessee Eastman process. Several enhanced autoencoder architectures exist to optimize the performance of such techniques including deep autoencoders, denoising autoencoders, and variational autoencoders (Spina et al., 2024). These designs enhance the feature learning capacity and robustness, resulting in higher fault detection rates and fewer false alarms compared to basic autoencoders.

Combined methods further enhance the fault detection ability. Combining autoencoders with LSTM networks offers an anomaly detection approach while also enabling classification through consideration of the process data's temporal dependency (Park et al., 2019). More recent research combines autoencoders and transformer networks for improved feature learning and fault detection (Shang & Yu, 2024). Despite the effectiveness of autoencoder-based fault detection, ample training data is required, and careful model design is crucial to prevent over-fitting; their performance also hinges critically on architecture and hyper-parameter choices. Autoencoders provide an effective and flexible tool for fault detection, particularly in nonlinear and high-dimensional industrial processes, and serve as a strong alternative to conventional PCA methods.

2.3 Comparison Between PCA and Autoencoder-Based Methods

The main difference between PCA and Autoencoder-based methods lies in their ability to model data complexity. PCA is a linear approach that extracts features by maximizing variance, and identifies faults through statistical indices like α and SPE (Gajjar et al., 2018). It is computationally inexpensive, interpretable, but limited for nonlinear and high-dimensional systems (Fan et al., 2017). Conversely, autoencoders are nonlinear neural networks capable of learning a latent space by minimizing reconstruction error, thereby encoding complex dependencies in process data and detecting subtle anomalies undetected by PCA (Fan et al., 2017). Results on the Tennessee Eastman process demonstrated higher detection rates using autoencoder-based methods than on PCA or similar models (Spina et al., 2024). Feature learning is another important distinction. PCA focuses on variance-driven components which might not be fault-relevant; conversely autoencoders discover features that facilitate accurate reconstruction and enhance fault sensitivity. For applicability, PCA excels at simple real-time applications due to low computational requirements. Autoencoders necessitate training on large data sets but provide better flexibility, with variants to handle time-series data (e.g., LSTM) or advanced structures (e.g., transformer-based networks) (Park et al., 2019). PCA excels at interpretability and simplicity, whereas autoencoders provide enhanced performance in nonlinear, dynamic, high-dimensional settings, positioning them better for modern industrial fault detection.

2.4 Research Gap

While PCA and Autoencoder-based methods have been extensively used in fault detection and diagnosis, there are still several limitations in the existing literature. First, most work individually analyzes PCA or Autoencoder-based methods. There is limited work providing a direct and unbiased comparison between these two methods under the same experimental setups. The traditional PCA-based methods are fairly well-known as well as applied across the industry, as they make them easy to interpret by simple linear models; however, the limitations concerning handling nonlinear, non-Gaussian, and time-varying data have long been discussed (Fan et al., n.d.; Li et al., 2000). Conversely, Autoencoder and deep learning algorithms performed much better at capturing complex patterns in data and improving fault detection accuracy. Yet, a large part of existing studies tend to concentrate on designing new models, and they do not focus on a direct, methodically comparable approach with traditional techniques in a similar environment (Yu & Zhang, 2023).

Moreover, there are few studies assessing the practical industrial relevance of both approaches in terms of computational efficacy, interpretability, and noise resistance, amongst others. Most research focuses on only linear statistical techniques or advanced deep learning methods, and no comprehensive examination is done of the strengths and weaknesses of the combined application under the integrated framework. This absence of thorough comparison results in unpredictability for practitioners in choosing appropriate FDD methods for respective industries. Hence, the present review seeks to fill these gaps by providing a structured comparison between PCA and Autoencoder-based models, emphasizing their performance, adaptability, and practical implementation in real-world industrial process monitoring.

3. Objectives

- To develop a data-driven fault detection framework using AutoEncoder-based deep learning for multivariate process data
- To implement PCA-based monitoring methods (Hotelling's T^2 and SPE) as a benchmark for comparison
- To evaluate and compare the performance of linear and nonlinear methods using FDR, FAR, ACC, and TTD
- To analyze the effectiveness of each method in detecting different types of faults in process systems.

4. Methodology

This project adopts a data-driven fault detection approach using a deep AutoEncoder model. The dataset used here is obtained from the sample dataset provided in the sample project by the course instructor and is derived from offshore gas processing plant simulation. The dataset contains multivariate time-series measurements (105 variables) representing process conditions such as pressure, temperature, and flow under both normal and faulty operating scenarios. Instead of developing a new physical model, this work utilizes the provided dataset and focuses on building a machine learning for fault detection.

4.1 Data Preprocessing and Split

The first step in the methodology is data preparation and preprocessing, implemented in Python. The dataset is divided into training and validation sets using a 97%–3% split, where the initial portion of the data is

used to train the model and the remaining portion is used for validation. This sequential split preserves the temporal nature of the process data. All variables are then normalized using z-score standardization through StandardScaler, ensuring that each variable has zero mean and unit variance. This step is essential because the dataset contains variables with different physical units and magnitudes, and normalization ensures stable and unbiased model training.

4.2 AutoEncoder Model

The core of the methodology is the development of a deep AutoEncoder (AE) using PyTorch. The AutoEncoder is designed with an encoder–decoder architecture, where the encoder compresses the high-dimensional input data into a lower-dimensional latent space, and the decoder reconstructs the original data from this compressed representation. The model structure consists of an input layer (105 features), two hidden layers, and a latent layer, followed by a symmetric decoder. The architecture ensures dimensionality reduction, enabling the model to learn the underlying structure of normal process behavior. The model is trained using the mean squared error (MSE) loss function, which minimizes the difference between the input and reconstructed output. To ensure optimal model performance, hyperparameter tuning is conducted using Optuna. Key parameters, including hidden layer sizes, latent dimension, learning rate, batch size, weight decay, and activation functions, are systematically optimized. The AutoEncoder is trained using the Adam optimizer during each trial. A structural constraint ($\text{hidden1} > \text{hidden2} > \text{latent}$) is imposed to maintain progressive dimensionality reduction.

Each trial trains a new model and evaluates it using validation reconstruction error, while inefficient trials are pruned early to improve computational efficiency. This automated tuning ensures that the final model is not arbitrarily selected but is systematically optimized for best performance. After training, the AutoEncoder learns the normal operating behavior of the process.

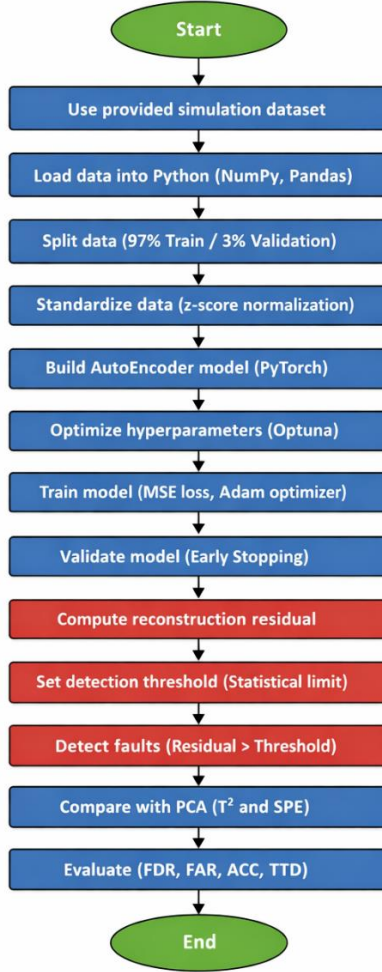


Figure 1:Flowchart of the proposed AutoEncoder-based fault detection.

4.3 Fault Detection Method

Fault detection is then performed using the reconstruction residual, which measures the deviation between actual and reconstructed data. For each sample, the reconstruction error is calculated, and a threshold is defined based on the statistical distribution of validation errors. If the residual exceeds the threshold, the condition is classified as a fault. This approach does not require labeled fault data during training, making it an unsupervised learning method. The effectiveness of the methodology is validated using the results obtained from the trained model. The AutoEncoder shows strong convergence during training, achieving a low validation loss of approximately 0.0029 and converging within 41 epochs. The model is then tested on different fault scenarios, including actuator faults, temperature faults, and pressure faults.

4.4 Benchmarking and Performance Evaluation

Performance is evaluated using metrics such as Fault Detection Rate (FDR), False Alarm Rate (FAR), Accuracy (ACC), and Time to Detect (TTD). The results demonstrate that the AutoEncoder achieves high detection accuracy (above 98%) and faster detection compared to traditional PCA-based methods. For benchmarking, PCA-based monitoring methods (Hotelling's T^2 and SPE) are also applied to the same dataset. The comparison shows that while PCA methods perform reasonably well for linear disturbances, they struggle with nonlinear faults. In contrast, the AutoEncoder provides more reliable and faster detection due to its ability to capture nonlinear relationships within the process data. We integrated the provided simulation dataset, developed machine learning model, automated optimization, and performance evaluation into a complete fault detection framework. The key advantage of our approach is that it leverages existing high-fidelity simulation data while introducing an advanced, nonlinear monitoring technique that significantly improves fault detection performance in complex process systems.

5. Results and Discussion

The training convergence behavior of the AutoEncoder model is illustrated in Figure 2. The training and validation losses decrease rapidly during the initial epochs, indicating that the model effectively captures the dominant nonlinear relationships within the process data. The training loss stabilizes around $\sim 10^{-3}$, while the validation loss converges to approximately 0.0029, with early stopping applied at epoch 41. The gap between training and validation curves remains small, suggesting that the model does not suffer from significant overfitting. This convergence indicates that the AutoEncoder has successfully learned the intrinsic correlations among multivariate process variables, including pressure, temperature, and flowrate interactions across different units (separators, compressors, etc.). The nonlinear nature of the plant dynamics makes deep learning models particularly suitable, as they can approximate complex mappings that linear methods such as PCA cannot fully capture.

The smooth convergence and low reconstruction error confirm that the learned latent space represents a compact manifold of normal operation. This is critical because fault detection relies on deviations from this manifold. The early stopping mechanism further ensures generalization, preventing the model from memorizing noise, which is essential in industrial environments where measurement noise is present.

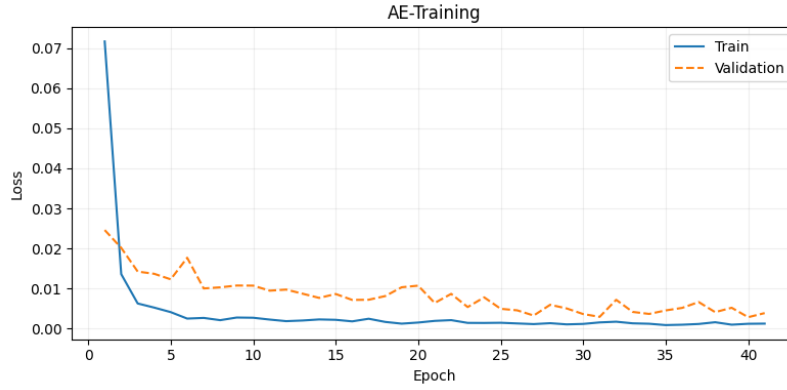


Figure 2: AutoEncoder Training Performance

5.1 Valve Fault (V-102 Fail-Hold)

The AutoEncoder effectively detects the valve fault with a high FDR (94.49%) and relatively fast detection (TTD = 4.32 min). The fail-hold condition causes loss of controllability, leading to nonlinear system-wide deviations, which are well captured by the AE. In contrast, PCA (T^2) performs poorly (FDR = 2.92%) due to its limitation in capturing nonlinear behavior. PCA (SPE) shows improved detection but with slightly slower response compared to AE.

Table 1: Fault Detection Performance – Valve Fault

Metric	AE	PCA (T^2)	PCA (SPE)
FDR (%)	94.49	2.92	96.27
FAR (%)	1.00	1.00	1.00
ACC (%)	95.74	29.62	97.03
TTD (min)	4.32	7.28	4.67

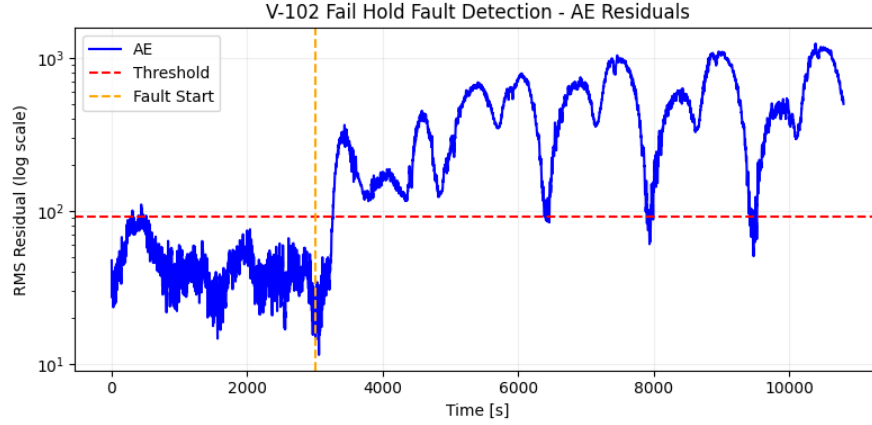


Figure 3: *AE Residuals – V-102 Fail Hold Fault*

Figure 3 presents the RMS reconstruction error of the AutoEncoder under a V-102 valve fail-hold fault condition. Before the fault occurrence (~ 3000 s), the residual remains below the threshold, indicating normal operation. After the fault is introduced, the residual exhibits a sharp increase and consistently exceeds the threshold. A fail-hold valve fault leads to loss of controllability, where the manipulated variable is fixed and cannot respond to disturbances. This results in process imbalance, causing deviations in flow distribution and pressure profiles across the separation unit. These deviations propagate through downstream units, leading to nonlinear system-wide effects. The AutoEncoder effectively detects this fault due to its sensitivity to nonlinear multivariate deviations. The oscillatory residual pattern observed after the fault suggests interaction between control loops attempting to compensate for the disturbance. The high FDR (94.49%) and low TTD (4.32 min) demonstrate that AE captures fault signatures earlier than traditional methods.

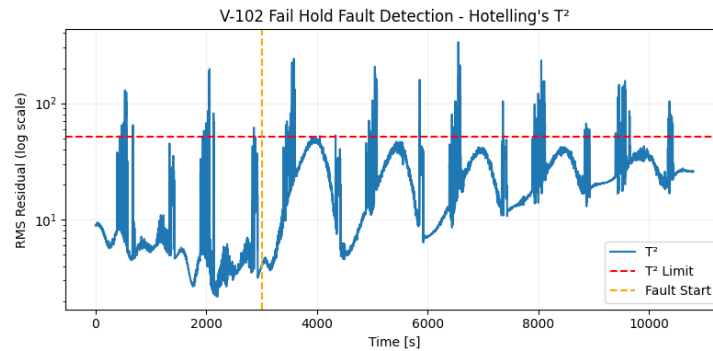


Figure 4: *PCA (T^2) – V-102 Fail Hold Fault*

Figure 4 shows the Hotelling's T^2 statistic under the same fault condition. Although some spikes exceed the control limit, the detection is inconsistent and delayed compared to AE. This behavior arises because PCA relies on linear correlations and variance structure. The fail-hold fault introduces nonlinear disturbances, which cannot be fully captured in the principal component space. As a result, the T^2 statistic fluctuates and fails to consistently exceed the threshold. Low FDR (2.92%) indicates poor sensitivity of PCA (T^2) to this type of actuator fault. This highlights a fundamental limitation of linear MSPC methods in complex process systems.

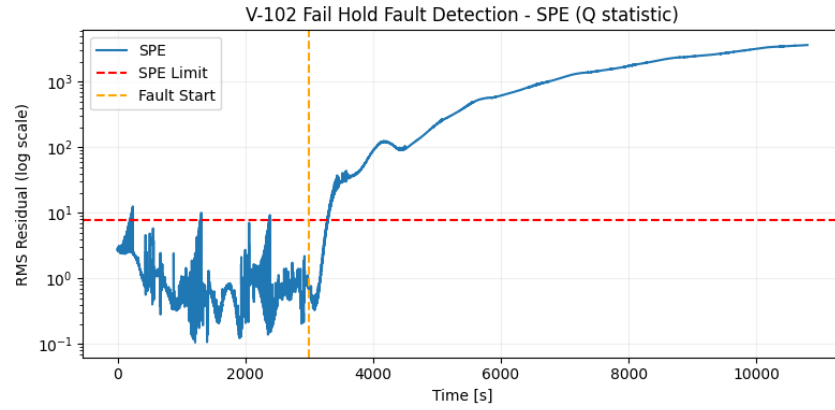


Figure 5: PCA (SPE) – V-102 Fail Hold Fault

Figure 5 illustrates the SPE (Q-statistic) response. After fault occurrence, the SPE increases significantly and remains above the threshold. Unlike T^2 , SPE captures residual variance (unmodeled behavior), making it more sensitive to faults that deviate from the PCA subspace. This explains the high FDR (96.27%). However, the gradual increase indicates slower detection dynamics, as SPE depends on accumulated reconstruction error in the residual space. Compared to AE, it lacks the ability to model nonlinear relationships, leading to slightly delayed detection.

5.2 Temperature Fault

The temperature fault is detected effectively by all methods due to strong system disturbances. The AutoEncoder achieves high accuracy and fast detection (TTD = 1.85 min). PCA (T^2) detects the fault but with slower response due to gradual variance changes. PCA (SPE) performs slightly better in detection speed, as it captures residual deviations outside the PCA subspace

Table 2: Fault Detection Performance – Temperature Fault

Metric	AE	PCA (T ²)	PCA (SPE)
FDR (%)	98.03	97.40	98.95
FAR (%)	1.00	1.00	1.00
ACC (%)	98.30	97.84	98.96
TTD (min)	1.85	3.38	1.07

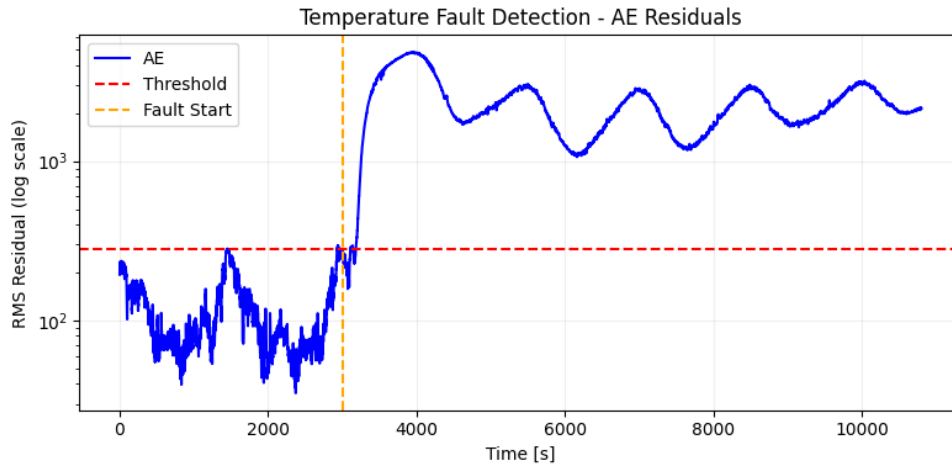


Figure 6: AE Residuals – Temperature Fault

Figure 6 shows AE residuals for a temperature disturbance fault. The residual increases sharply at the fault onset and remains consistently above the threshold. Temperature disturbances affect thermodynamic equilibrium, altering phase behavior and separation efficiency. This leads to cascading effects in pressure and flow across the plant. The AE captures these nonlinear interactions effectively. The smooth yet elevated residual pattern suggests stable but shifted operating conditions, typical of disturbance faults. The high FDR (98.03%) and fast TTD (1.85 min) confirm the robustness of AE in detecting process disturbances.

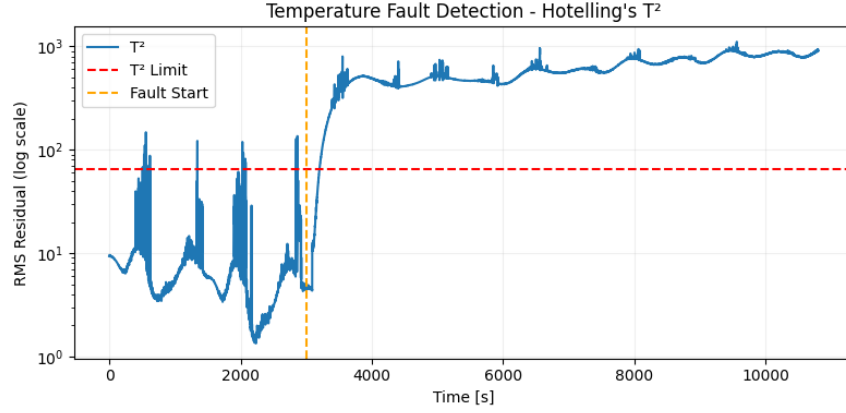


Figure 7: Temperature Fault Detection using Hotelling's T^2

Figure 7 illustrates the temporal response of Hotelling's T^2 statistic under a temperature fault scenario. The fault is introduced at approximately 3000 s, after which a significant increase in the T^2 values is observed, exceeding the predefined control limit. Prior to the fault occurrence, intermittent fluctuations are visible, indicating the sensitivity of the statistic to inherent process variability and measurement noise. From a process systems perspective, temperature disturbances affect multiple interconnected variables, including vapor-liquid equilibrium, fluid density, and separator efficiency. These changes induce variations in pressure and flow dynamics across the plant. Since Hotelling's T^2 measures the variation of process variables within the principal component subspace, it is primarily sensitive to changes in overall variance. However, due to the nonlinear interactions among process variables, particularly in thermodynamic systems, the linear PCA framework cannot fully capture the evolving system behavior.

A comparison with the AutoEncoder-based residuals (Figure 6) reveals that the T^2 statistic exhibits a relatively slower and more oscillatory response following fault initiation. While the AutoEncoder residual demonstrates a rapid and stable deviation from normal operation, the T^2 statistic gradually stabilizes above the threshold. This difference highlights the limitation of PCA in modeling nonlinear process dynamics. The observed detection delay ($TTD = 3.38$ min) compared to the AutoEncoder ($TTD = 1.85$ min) further confirms that nonlinear feature extraction methods provide superior fault detection performance in complex process systems.

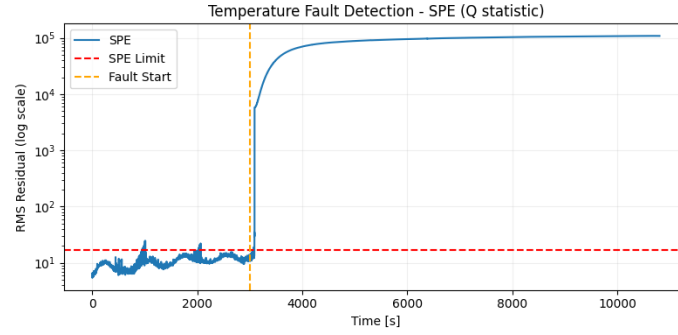


Figure 8: PCA (SPE) – Temperature Fault

Figure 8 shows a sharp increase in SPE immediately after fault occurrence. The rapid rise indicates that the temperature fault introduces significant model mismatch, pushing the system outside the PCA subspace. SPE performs better than T^2 due to its sensitivity to residuals.

5.3 Pressure Fault

All methods perform well for pressure faults due to large variance changes in the system. The AutoEncoder achieves perfect detection (100% FDR) with instantaneous response. PCA methods also perform well, as pressure disturbances strongly affect system variance, making them easier to detect even with linear models. The AutoEncoder consistently outperforms PCA-based methods, particularly for nonlinear faults such as valve failures. While PCA (T^2) struggles with nonlinear relationships, PCA (SPE) performs better but still lacks robustness compared to AE. The AE provides a superior balance of accuracy, robustness, and detection speed.

Table 3: Fault Detection Performance – Pressure Fault

Metric	AE	PCA (T^2)	PCA (SPE)
FDR (%)	100.00	92.56	100.00
FAR (%)	1.03	1.00	1.00
ACC (%)	99.71	94.35	99.72
TTD (min)	0.00	0.00	0.00

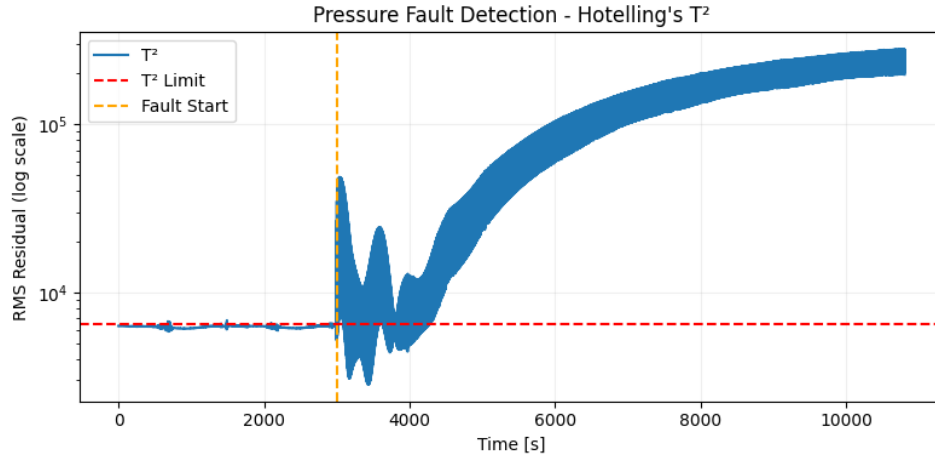


Figure 9: PCA (T^2) – Pressure Fault

Figure 9 shows the T^2 statistic under a pressure fault. A sharp increase is observed immediately after fault introduction. Pressure faults directly impact mass balance and flow distribution, causing large-scale variance changes in the system. Since PCA captures variance, it performs better here than in previous cases. However, the spread of values indicates instability and nonlinear propagation, which PCA cannot fully represent.

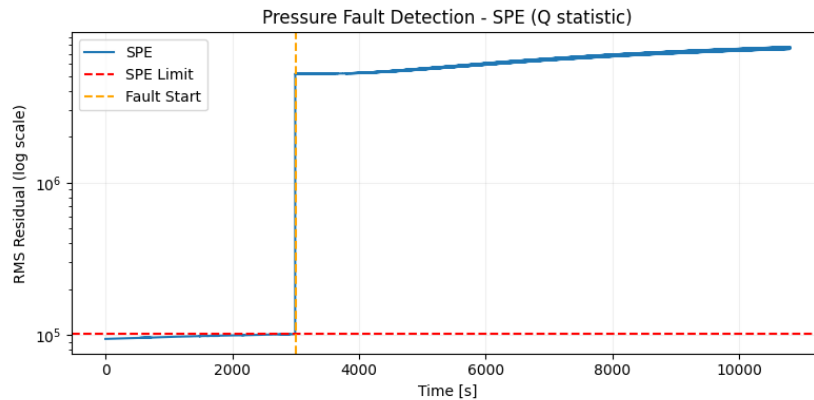


Figure 10: PCA (SPE) – Pressure Fault

Figure 10 shows SPE for the pressure fault, with a strong and immediate deviation beyond the threshold. This indicates that pressure faults create significant deviations from the PCA model, resulting in high residual energy. The SPE captures this effectively, achieving 100% FDR.

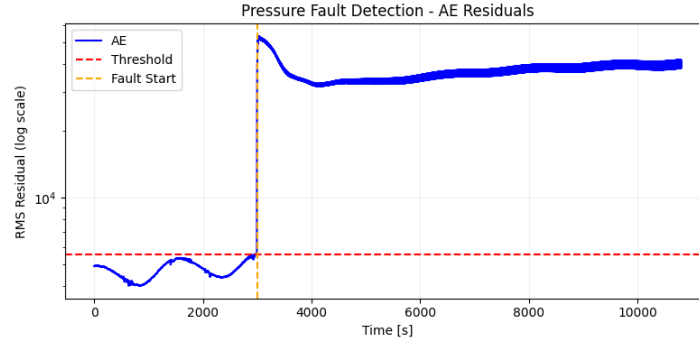


Figure 11: *AE Residuals – Pressure Fault*

Figure 11 illustrates AE residuals for the pressure fault. The residual increases sharply at the fault onset and remains significantly above the threshold. From a process perspective, pressure disturbances propagate rapidly across interconnected units, leading to strong nonlinear interactions. The AutoEncoder captures these interactions efficiently, resulting in instant detection (TTD = 0 min) and near-perfect accuracy. The stable elevated residual suggests that the system transitions to a new operating regime, rather than oscillatory instability. This highlights the ability of AE to distinguish between different types of faults.

5.4 Comparative Analysis

The comparative evaluation of AE and PCA-based methods (T^2 and SPE) demonstrates a performance differences across fault scenarios. The AE consistently achieves higher Fault Detection Rate (FDR) and faster Time to Detect (TTD), particularly for nonlinear faults such as the V-102 valve fail-hold condition. As shown in Figure 3, the AE residual exhibits a rapid and sustained deviation beyond the threshold immediately after fault initiation, indicating strong sensitivity to nonlinear multivariate changes. In contrast, Figure 4 shows that the PCA T^2 statistic responds inconsistently, with delayed and intermittent threshold violations, resulting in significantly lower detection sensitivity.

The SPE-based PCA method (Figure 5) improves detection performance by capturing residual variance outside the principal component subspace; however, its response remains comparatively slower due to gradual accumulation of reconstruction error. For temperature faults (Figures 6–8), all methods demonstrate effective detection due to strong system disturbances, although AE maintains a more stable and faster response. In pressure faults (Figures 9–11), all techniques achieve near-perfect detection due to dominant variance changes; however, AE still provides a more stable residual profile, indicating better robustness. These results confirm that AE outperforms PCA in capturing nonlinear relationships and

enabling early fault detection, while PCA remains effective for faults dominated by large linear variance changes.

Table 4:Comparative Analysis

Metric	AE	PCA (T^2)	PCA (SPE)
FDR	High	Low–Moderate	High
FAR	Low	Low	Low
TTD	Fast	Slow	Moderate
Nonlinearity Handling	Excellent	Poor	Moderate

The comparative analysis between the AutoEncoder (AE), PCA–Hotelling’s T^2 , and PCA–SPE methods highlights the superiority of nonlinear, data-driven approaches for fault detection in complex process systems. The AutoEncoder consistently achieves higher fault detection rates (FDR) and faster detection times (TTD) across all fault scenarios due to its ability to learn nonlinear relationships among process variables. In contrast, the PCA-based T^2 statistic, which relies on variance within a linear subspace, performs poorly for faults that introduce nonlinear disturbances, such as actuator failures, resulting in low detection sensitivity and delayed response. The PCA–SPE metric performs better than T^2 because it captures residual deviations outside the PCA model; however, it still lacks the capability to fully represent nonlinear process dynamics. While all methods maintain low false alarm rates (FAR), the AutoEncoder provides the best overall balance of accuracy, robustness, and speed. This demonstrates that nonlinear feature extraction is essential for reliable fault detection in highly coupled and dynamic industrial systems.

5.5 Key Observations

The results clearly indicate that the AE provides superior fault detection performance compared to PCA-based methods, particularly in complex and nonlinear process conditions. The AE effectively captures nonlinear interactions among process variables, enabling consistent and early detection of faults, as evidenced by its higher Fault Detection Rate (FDR) and lower Time to Detect (TTD). In contrast, the PCA (T^2) statistic, which relies on linear variance structures, shows poor sensitivity to nonlinear disturbances, resulting in delayed or inconsistent detection. Although the SPE statistic improves detection by capturing residual deviations outside the PCA subspace, it still exhibits slower response compared to the AE.

Furthermore, the results demonstrate that all methods perform well for faults associated with significant variance changes, such as pressure faults; however, the AE maintains a more stable and robust detection profile across all scenarios. Overall, these results highlight the importance of nonlinear feature extraction in modern process monitoring and confirm that AutoEncoder-based approaches are more suitable for detecting subtle and complex faults in industrial systems.

6. Limitations

The proposed AutoEncoder-based framework shows strong fault detection performance; however, it has key limitations. The primary limitation is the lack of temporal modeling, as the AutoEncoder treats each sample independently. As seen in Figures 3, 4, and 7, fault signatures evolve over time, showing gradual deviations and oscillatory behavior after fault occurrence. The AutoEncoder treats each data sample independently, without considering time dependencies between consecutive measurements. From a process control perspective, industrial systems are inherently dynamic, and faults often propagate through units with time delays and feedback interactions. This limitation can result in delayed detection (as seen in PCA T^2) or oscillatory residual patterns (in AE), reducing the reliability of early fault identification. To address this, incorporating deep learning models such as LSTM-AutoEncoders or sequence-based architectures would allow the model to learn time-dependent relationships, significantly improving detection accuracy and responsiveness. Additionally, the model shows sensitivity to noise, where small fluctuations produce minor spikes in residuals under normal operation. This may increase false alarms in practical applications. These issues can be improved through signal smoothing, hybrid AE-PCA frameworks, adaptive thresholding, explainable AI techniques, and inclusion of more diverse datasets to enhance robustness and practical applicability.

7. Conclusion

This project presented a data-driven framework for fault detection in process systems using AutoEncoder-based deep learning and PCA-based statistical monitoring. The results demonstrate that the AutoEncoder provides superior performance in terms of detection accuracy, robustness, and speed, particularly for nonlinear and complex fault scenarios. While PCA-based methods remain effective for faults characterized by significant variance changes, their limitations in modeling nonlinear relationships reduce their reliability

in modern industrial systems. The findings highlight the importance of nonlinear feature extraction techniques for effective process monitoring. The AutoEncoder's ability to learn intrinsic system behavior enables early and consistent fault detection, making it a promising tool for industrial applications where safety and reliability are critical.

8. Future Work

Future work should focus on enhancing the proposed framework by incorporating temporal deep learning models, such as LSTM-AutoEncoders and transformer-based architectures, to capture dynamic process behavior more effectively. Additionally, the integration of denoising techniques and adaptive thresholding can improve robustness against measurement noise. Further research should also explore explainable AI (XAI) methods to improve interpretability of AutoEncoder-based fault detection, addressing one of its key limitations. Finally, validation on real industrial datasets and consideration of real-time implementation constraints will be essential to ensure scalability and practical applicability.

References

- Venkatasubramanian, V., Rengaswamy, R., & Yin, K. (2003). A review of process fault detection and diagnosis: Part I – Quantitative model-based methods. *Computers & Chemical Engineering*, 27(3), 293–311.
- Venkatasubramanian, V., Rengaswamy, R., Kavuri, S., & Yin, K. (2003). A review of process fault detection and diagnosis: Part III – Process history based methods. *Computers & Chemical Engineering*, 27(3), 327–346.
- MacGregor, J. F., & Kourti, T. (1995). Statistical process control of multivariate processes. *Control Engineering Practice*, 3(3), 403–414.
- Wold, S., Esbensen, K., & Geladi, P. (1987). Principal component analysis. *Chemometrics and Intelligent Laboratory Systems*, 2, 37–52.
- Hinton, G. E., & Salakhutdinov, R. R. (2006). Reducing the dimensionality of data with neural networks. *Science*, 313(5786), 504–507.
- Ge, Z., Song, Z., Ding, S. X., & Huang, B. (2017). Data mining and analytics in the process industry: The role of machine learning. *IEEE Access*, 5, 20590–20616.
- Fan, J., Wang, W., & Zhang, H. (2017). AutoEncoder based high-dimensional data fault detection system. *2017 IEEE 15th International Conference on Industrial Informatics (Indin)*, 1001–1006.
- Gajjar, S., Kulahci, M., & Palazoglu, A. (2018). Real-time fault detection and diagnosis using sparse principal component analysis. *Journal of Process Control*, 67, 112–128.
- Park, P., Marco, P. D., Shin, H., & Bang, J. (2019). Fault detection and diagnosis using combined autoencoder and long short-term memory network. *Sensors*, 19(21), 4612.
- Pozo, F., & Vidal, Y. (2015). Wind turbine fault detection through principal component analysis and statistical hypothesis testing. *Energies*, 9(1), 3.
- Shang, J., & Yu, J. (2024). A residual autoencoder-based transformer for fault detection of multivariate processes. *Applied Soft Computing*, 163, 111896.
- Spina, D. E., Campos, L. F. de O., de Arruda, W. F., Melo, A., Alves, M. F. de S., Rabello, G. L., Anzai, T. K., & Pinto, J. C. (2024). Comparison of autoencoder architectures for fault detection in industrial processes. *Digital Chemical Engineering*, 12, 100162.
- Tamura, M., & Tsujita, S. (2007). A study on the number of principal components and sensitivity of fault detection using PCA. *Computers & Chemical Engineering*, 31(9), 1035–1046.
- Villegas, T., Fuente, M. J., & Rodríguez, M. (2010). Principal component analysis for fault detection and diagnosis. Experience with a pilot plant. *Advances in Computational Intelligence, Man-Machine Systems and Cybernetics*, 147–152.

Wang, S., & Xiao, F. (2004). AHU sensor fault diagnosis using principal component analysis method. *Energy and Buildings*, 36(2), 147–160.

Deng, X., Tian, X., Chen, S., & Harris, C. J. (2017). Deep learning based nonlinear principal component analysis for industrial process fault detection. *IEEE Conference Paper*.

Fan, J., Wang, W., & Zhang, H. (n.d.). Autoencoder based high-dimensional data fault detection system. *IEEE Conference Paper*.

Gajjar, S., Kulahci, M., & Palazoglu, A. (2018). Real-time fault detection and diagnosis using sparse principal component analysis. *Journal of Process Control*, 67, 112–128.

Khaled, M. S., Imtiaz, S., Ahmed, S., & Zendejboudi, S. (2021). Dynamic simulation of offshore gas processing plant for normal and abnormal operations. *Chemical Engineering Science*, 230, 116159.

Khaled, M. S., Imtiaz, S., Ahmed, S., & Zendejboudi, S. (n.d.). Fault detection and diagnosis using combined principal component analysis and artificial neural network. *Conference/Manuscript*.

Li, W., Yue, H. H., Valle-Cervantes, S., & Qin, S. J. (2000). Recursive PCA for adaptive process monitoring. *Journal of Process Control*, 10(5), 471–486.

Wang, Y., Pan, Z., Yuan, X., Yang, C., & Gui, W. (2020). A novel deep learning based fault diagnosis approach for chemical process with extended deep belief network. *ISA Transactions*, 96, 457–467.

Yu, J., & Zhang, Y. (2023). Challenges and opportunities of deep learning-based process fault detection and diagnosis: A review. *Neural Computing and Applications*, 35, 211–252.

Lee, J.-M., Yoo, C., & Lee, I.-B. (2004). Statistical process monitoring with independent component analysis. *Journal of Process Control*, 14(5), 467–485.

Iqbal, R., Maniak, T., Doctor, F., & Karyotis, C. (2019). Fault detection and isolation in industrial processes using deep learning approaches. *IEEE Transactions on Industrial Informatics*.